inClass-L36

Monday, October 26, 2026 6:50 AM

# CV2090
# Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

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## Announcement

# Course Flow

Module 1: **Probability** – Understanding Uncertainty

Foundations of Probability → RVs & Probability Models → Multivariate Probability → Modeling Extremes

Module 2: **Statistics** – Learning from Data

Descriptive statistics → Statistical Inference → Regression & Statistical Learning

Module 3: **Reliability** – Decision under Uncertainty

Reliability Analysis → Computational Reliability → Stochastic Processes & Time-Dependent Reliability → Risk-Informed Decision Making

# Why should you care about Risk & Reliability?

- Space Shuttle *Challenger* disaster (1986)
  https://www.youtube.com/watch?v=yibNEcn-4yQ





Catch-all approach:

Build for every "what if" scenario: create backup for every imaginable scenario, even wildly unlikely ones

**What could go wrong with this approach?**

## Why should you care about Risk & Reliability?





**Catch-all approach:**

**Build for every "what if" scenario:** create backup for every imaginable scenario, even wildly unlikely ones

Spreads engineers too thin: divided resources, real threats do not get the focus they need

Illusion of safety: looks thorough on paper, but critical risks can slip through because everything is treated as equally important

IF EVERYTHING IS A PRIORITY, THEN NOTHING IS.

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# Typical Probabilities

# Typical Probabilities

- Some target probabilities are:
  - Locally dramatic but rather harmless in a wider area
    e.g., a building collapse $\leq 10^{-4}$/year
  - More dramatic failure consequences, such as nuclear power plants
    e.g., a nuclear power plant – Meltdown $\leq 10^{-5}$/year
    – Leakage $\leq 10^{-6}$/year
  - Drawing an ace of spade in a randomized deck of cards
    $1/52 \approx 2\times10^{-2}$
  - Rule of thumb,
    Probability of death/person/year $> 10^{-3} \Rightarrow$ Immediate Action
    $< 10^{-6} \Rightarrow$ Events so unusual that little reasonable can be done about it.



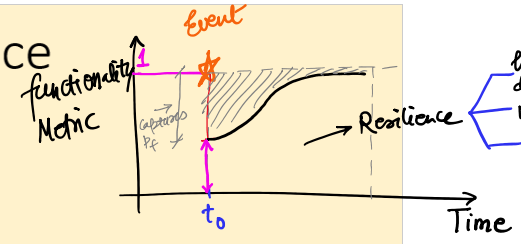
# Typical Probabilities

- Rule of thumb, Probability of death/person/year
  - $> 10^{-3} \Rightarrow$ Immediate Action
  - $< 10^{-6} \Rightarrow$ events so unusual that little can reasonably be done
- Examples:
  - Worker in unregulated mine in early 20th century
    $\sim 1/100$
  - Riding a motorbike without a helmet (in low-income countries)
    $\sim 10^{-4}$
  - Riding a motorbike with a helmet (in low-income countries)
    $\sim 10^{-5}$
  - Structural collapse of old and ill-maintained buildings
  - Commercial plane $1.2\times10^{-8}$/person-hour $\Rightarrow$ $24\times10^{-6}$
  - Car $0.7\times10^{-8}$/person-hour $\Rightarrow$ $200\times10^{-6}$
    (accounting for average exposure)



# Risk, reliability, resilience

- $p_f$: Pr(failure)
- Risk: $P_f$
- Reliability: $1 - P_f$
- Resilience: Response in the aftermath




How low do you go in immediate aftermath → Robustness
Where do you bounce back? ≡ Total loss → RES → Recovery
How quickly do you bounce back? → Rapidity

- Resilience: Response in the aftermath of an event

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## Definition of Failure (Limit State)

How to define 'failure'

Failure is system performs worse than your expectation.

Limit state
- Demand, S
- Capacity, R

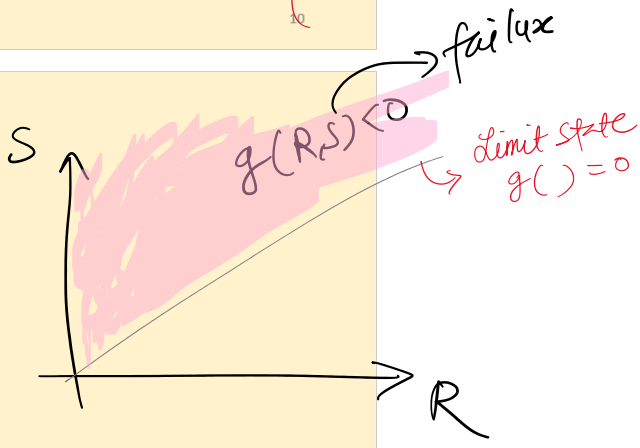
Boundary between 'what's safe/ acceptable' and 'what's NOT safe/ acceptable'

↑ Limit state function, $g(R,S) = R - S$

- $<0$ failure
- $>0$ safe

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## Limit State Function




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